Pathfinder Test Procedure Document

1. Introduction

This Test Procedure Document outlines the step-by-step process for verifying the functionality of the Pathfinder web application. The goal is to ensure that all core features of Pathfinder work as expected, fulfilling the system's requirements as described in the Software Requirements Specification (SRS). The test procedure covers unit, integration, and acceptance testing to validate individual components and the fully integrated system.

2. Test Execution Process

The test procedure follows these steps:

1. Start with the list of functional requirements and cross-reference with the test cases.
2. Start the Pathfinder software.
3. Perform individual tests for each feature, ensuring all acceptance criteria are met.
4. Run tests for integrated components to verify smooth interaction between modules.
5. Execute full-system tests to confirm Pathfinder’s overall functionality.
6. Document test results and address any failures or discrepancies.

3. Test Cases

3.1 Unit Test Procedures

These tests verify that individual modules function correctly before integration.

UT1: User Login Test

Objective: Ensure users can log in with valid credentials.

1. Open the Pathfinder application.
2. Navigate to the login page.
3. Enter valid credentials (email and password).
4. Click "Sign In."
5. Verify the user is redirected to the dashboard.
6. Log out and verify that the session terminates correctly.

UT2: Resume Upload Test

Objective: Confirm users can upload a resume file and receive AI feedback.

1. Log in to Pathfinder.
2. Navigate to the Resume Analysis page.
3. Click "Upload Resume" and select a valid resume file.
4. Verify that the AI provides analysis results.

UT3: Job Search Test

Objective: Ensure job search functionality works with user preferences.

1. Log in to Pathfinder.
2. Navigate to the Job Matching page.
3. Enter job preferences (location, salary, remote option).
4. Click "Search" and verify job listings match the criteria.

3.2 Integration Test Procedures

These tests confirm that integrated components interact correctly.

IT1: Login & Dashboard Integration Test

Objective: Ensure seamless transition from login to dashboard.

1. Log in to Pathfinder with valid credentials.
2. Verify that the dashboard loads with user-specific data.

IT2: Resume-Based Job Matching Test

Objective: Confirm that uploaded resumes influence job recommendations.

1. Upload a resume.
2. Navigate to the Job Matching page.
3. Verify that job listings are tailored based on resume content.

IT3: AI Chatbot Career Advice Test

Objective: Ensure the chatbot provides relevant career suggestions.

1. Open the chatbot window.
2. Enter a career-related query (e.g., "What jobs fit my skills?").
3. Verify that the chatbot provides a reasonable response.

3.3 System Acceptance Test Procedures

These tests confirm that Pathfinder meets all acceptance criteria.

AT1: User Journey Test

Objective: Ensure a user can successfully complete core actions.

1. Create a new user account.
2. Upload a resume.
3. Search for jobs.
4. Ask the chatbot for career guidance.
5. Log out and verify account data remains saved.

AT2: Notification System Test

Objective: Confirm users receive notifications for relevant job alerts.

1. Enable job alerts in user settings.
2. Perform a job search.
3. Log out and wait for notification.
4. Verify that Pathfinder sends a job alert notification.

4. Test Reporting

All test results will be documented, including:

- Pass/Fail status.
- Errors encountered.
- Screenshots (if applicable).
- Suggestions for fixes and improvements.

5. Conclusion

By following this test procedure, Pathfinder's functionality will be thoroughly validated to ensure a seamless user experience. Any failed tests will be logged and corrected before deployment.